Learning Statistically Accurate Dynamics of Complex Systems

Tan Bui†, Cody McKenna†, Singer Xing†, Daniel Youngberg†

†University of Wisconsin - Madison, equal contribution

Introduction

Dynamical systems model many important real world phenomena such as the weather outside, the neurons in your brain, and so much more. To model really complex dynamical systems, we have a set of models called operational models. These can be super expressive, capable of modeling complicated relationships. However, they can get caught up in complexity and fail to model some of the most basic statistical quantities, such as the average temperature of an area. We propose a latent data assimilation framework which can model complex dynamics and uses statistical measurements to keep its predictions grounded.

Data Assimilation and EnKF

1. **What is Data Assimilation?**
   - *Goal:* Improve forecasts by combining a dynamic model with current and historical noisy observations to produce an optimal estimate of the true system.
   - *Kalman filter:* A recursive two-step pipeline where the model makes a prediction *(forecast)* and then corrects itself based on past observations *(analysis)*.

$$p(\mathbf{x}_{k-1} \mid Z_{k-1}) \xrightarrow{\text{Forecast}} p(\mathbf{x}_k \mid Z_{k-1}) \xrightarrow[p(\mathbf{z}_k|\mathbf{x}_k)]{\text{Analysis}} p(\mathbf{x}_k \mid Z_k)$$

   - *Bayesian inference:*

$$p(\mathbf{x}_k \mid \mathbf{z}_k) \propto p(\mathbf{x}_k) p(\mathbf{z}_k \mid \mathbf{x}_k)$$

2. **From Kalman Filter to EnKF**
   - *Classical Kalman filter* (linear/Gaussian): propagate state mean and covariance; analysis via Kalman gain.
   - *Ensemble Kalman Filter [2]* (nonlinear): represent uncertainty via ensemble members; replace covariance matrices by sample covariances.

Latent Dynamics

1. **Distributional Observation**

✗ *Challenge:* In certain scenarios, we only receive access to statistical moments of physical states (with observation noise):

$$\mathbf{m} = \mathcal{M}(\mathbf{x}) + \boldsymbol{\varepsilon}.$$

The inverse problem is inherently ill-posed: as $\mathcal{M}$ is many-to-one, the physical state cannot be uniquely recovered from observational moments alone.

2. **Latent Space Representation**

✗ *Challenge:*
   - ○ High-dimensional data assimilation suffers from *curse of dimensionality*.
   - ○ Standard autoencoders can reduce dimensionality but lack physical interpretability, which prevents integration of physics-based insights.

- *Latent space:* Augment the latent vector generated by an encoder $\mathcal{E}$ with statistical measurements to align the latent space with key physical features [1]:

$$\mathbf{x}_k = \begin{bmatrix} \mathcal{E}(\mathbf{X}_k) \\ \mathbf{m}_k \end{bmatrix}.$$

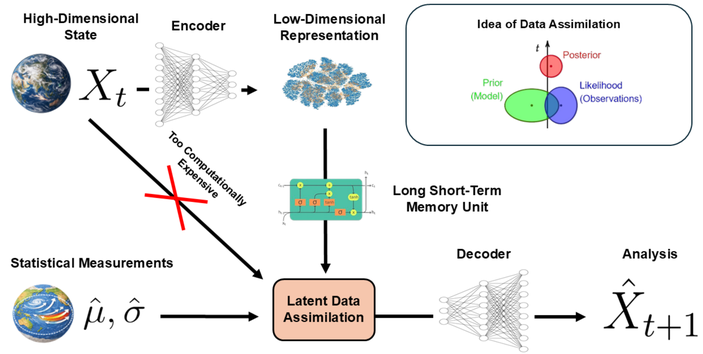
- *Architecture:* We train a CNN autoencoder that maps a correspondence between physical states and their low-dimensional latent variables:

$$\mathbf{z} = \mathcal{E}(\mathbf{x}), \quad \hat{\mathbf{x}} = \mathcal{D}(\mathbf{z}).$$

The decoder $\mathcal{D}$ acts as a **closure** that "fills in" unresolved degrees of freedom when translating latent variables to full state realizations.

Methodology Overview

The schematic diagram below illustrates the workflow of the proposed framework:



Physical Models

We used two popular stochastic dynamical models to generate our training data:

- **Lorenz 96**: With the conditions $F = 5$, $\sigma = 0.5$, $N = 40$, the model follows the SDE:

$$\mathrm{d}x_n = (x_{n-1}(x_{n+1} - x_{n-2}) - x_n + F)\,\mathrm{d}t + \sigma\,\mathrm{d}W_n(t), \qquad n = 1, \ldots, N.$$

- **FitzHugh-Nagumo (FHN)**: With the conditions

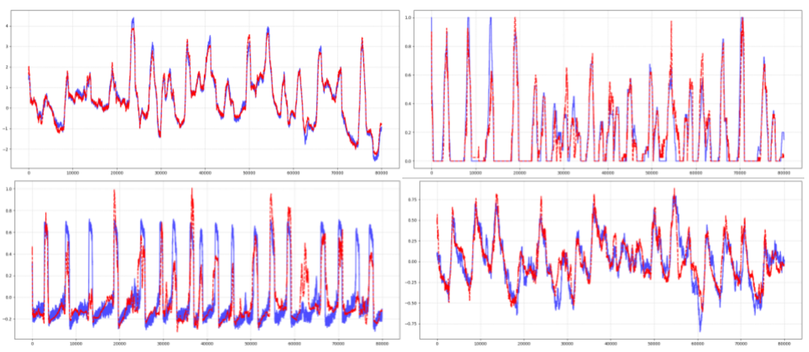
$$N = 40,\ \epsilon = 10^{-2},\ d_u = 0.5,\ c = \frac{1}{3},\ c_1 = 1,\ \delta_1 = 0.2,\ a = 1.05,\ c_2 = 0,\ \delta_2 = 0.4,$$

the model follows the coupled SDE:

$$\begin{aligned} \epsilon\,\mathrm{d}u_n &= (d_u(u_{n-1} - 2u_n + u_{n+1}) + u_n - cu_n^3 - c_1 v_n)\mathrm{d}t + \sqrt{\epsilon}\delta_1 \mathrm{d}W_{u,n}(t), \\ \mathrm{d}v_n &= (u_n + a - c_2 v_n)\mathrm{d}t + \delta_2 \mathrm{d}W_{v,n}(t), \qquad n = 1, \ldots, N. \end{aligned}$$

For each model, we generate training data using the Euler-Maruyama numerical integration scheme with $T = 80000$ time points.

Results



FHN LSTM Results: Red is reconstructed, Blue is truth. Top-left: latent space, Top-right: percentage of u above a spike threshold, Bottom-left: one dimension of u, Bottom-right: one dimension of v.

Discussion

- To apply our framework to the Lorenz '96 model, we directly feed our autoencoder the 40-dimensional state vector. From there, we encode it to 8 dimensions, where we append 5 physical measurements to bring the encoded dimension to 13 dimensions. Finally, we perform our data assimilation on these 13-dimensional vectors.
- To apply our framework to the FitzHugh-Nagumo model, we concatenated two variables with 40 dimensions to end up with 80-dimensional training data, which we still trained an autoencoder to transform into an 8-dimensional space.
- We trained the LSTM on the latent space with four different statistical observations added. These were u-v cross correlation, percentage of u's spiking above a set threshold (visualized in the top right figure), and the mean of both u and v.

Conclusion and Future Work

- Next step is to fully implement EnKF. Our presented data is produced by an augmented latent space trained LSTM.
- Other statistical observations that may be considered for the FHN system include the variance of u and v and their lagged cross-correlation statistics as a statistical indicator of the u and v causality.
- For the Lorenz '96 model, we will augment the latent space with different statistical measurements, such as Fourier modes and high-order moments, and then assimilate the corresponding data into the LSTM model. This allows us to move beyond traditional assimilation techniques that focus only on pathwise recovery of the physical states.

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References

[1] Pouria Behnoudfar et al. "Bridging idealized and operational models: an explainable AI framework for Earth system emulators". In: *npj Climate and Atmospheric Science* (2026).

[2] Geir Evensen. "The ensemble Kalman filter: Theoretical formulation and practical implementation". In: *Ocean dynamics* 53 (2003), pp. 343–367.